DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2021/2022

SECOND YEAR **SEMESTER II** EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CIT 2205: MOBILE COMPUTING

DATE: **TIME:**

Instructions: Answer Question 1 and Any Other Two

Question 1: (30 Marks)

a) Discuss the technological advancements between the second (2G) and the third-generation (3 G) mobile communication. (2 marks)
b) Describe Four WAP site design tips you should keep in mind. (4 marks)
c) Explain three reasons for a handoff to be conducted in mobility management (3 marks)
d) Using a diagram, describe the IP mobile registration process (4 marks)

e) Mobile communication is resource constrained. Therefore, it requires special operating environment which translates TCP/IP protocols into an equivalent of wireless applications environment. Discuss how the WAP architecture seamlessly integrates www internet applications (4 marks)
f) Assume that your friend recently travelled to West Africa and you just made a phone call to her. She is surprised and is asking how her Safaricom Kenya line worked in Nigeria. Discuss mobile IP topology and explain how it facilitates such kind mobility management beyond a particular location area (4 marks)
g) Explain any three problems associated with Mobile IP (3 marks)
h) One of the key challenges in mobile computing is mobility, users keep changing locations now and then, hence, the need for location management. Discuss three strategies that are used in location management based on pure paging (6 marks)

Question 2: (20 Marks)

a) Explain the two basic operations in roaming management: (4 marks)
b) In mobile computing, future developments will definitely provide new technologies and better computing concepts. Explain any TWO of issues being addressed by these (2 marks)

c) Explain how Frequency reuse help channel allocation of cellular systems (4 marks)

d) Explain three approaches that have been proposed to reduce registration traffic during call delivery process (6 marks)

e) Differentiate between soft handoff and hard handoff (4 marks)

f) Discuss the components of personal communication service (PCS) architecture (2 marks)

Question 3: (20 Marks)

a) Define the following terms as used in mobile computing (3 marks)
   i) Network mobility
   ii) Terminal mobility
   iii) polling

b) Explains the improvements of IPV6 that has solved the challenges raised in IPV4. (3 marks)

c) To reduce force termination and promote call completion channel assignment scheme have been proposed. Using illustrations discuss four channel assignment schemes that have been proposed (8 marks)

d) Describe in detail the following occurrence of media access in mobile computing. Use diagram where necessary.
   i. Exposed Terminals (2 marks)
   ii. Hidden Terminal (2 marks)
   iii. Near and Far Terminal (2 marks)

Question 4: (20 Marks)

a) Differentiate between *mobility* and *wireless* terminologies as used in mobile computing. (2 marks)

b) Interference in the channels is a major challenge in wireless communication. To address this challenge multiplexing mechanism can be used. Explain four dimensions in which multiplexing can be carried out in wireless communication. (8 marks)

c) Onyango Travelled to France for a holiday. While in France Onyango was able to check balance on his mobile phone, he could also receive calls and short messages as well as send and call his family back home in Kenya using his Safaricom Line. This can be explained using roaming management in mobility; thus, one is able to make calls or even receive messages while outside his or her geographical region. Using a diagram discuss the three key components of signaling system 7 (SS7) network that necessitated Onyango to make and receive calls and messages while in France. (6 marks)

d) Explain the following handover types use illustration where possible (4 marks)
   i) Inter-cell handoff
   ii) intersystem handoff

END.